Role 3 — Week 4: Comparative Statics and Robustness

Team A1

1 Signed characterization comparative statics

Let $J(\theta) := \text{Carbon}_{\text{FCL}}(\theta) - \text{Carbon}_{\text{no-FCL}}(\theta)$ be the carbon functional at the joint optimum. Write $x = D - K_R\eta$, let $F(y; K_R)$ be the residual CDF, and reserve $g(x) = F_{K_R}(x)/f(x)$ for the shift ratio (as in Roles 1–2). The FCL first-order conditions (eqs. 12–13, Week 1) fix $F(K_F^*) = \bar{q}_F$ and $F(K_F^* - K_M^*) = \bar{q}_{F-M}$, where $\bar{q}_F = (\beta + p - c - C_F - C_M)/(\beta + p - R)$ and $\bar{q}_{F-M} = C_M/(R - c)$.

Proposition 1 (Signs of $\partial J/\partial\theta$)**.** *Under Assumption 1 and at any interior joint optimum with strictly positive K_M^*,*

$$\text{sign}\left(\frac{\partial J}{\partial \theta}\right) = \begin{cases} +, & \theta = R, \\ -, & \theta \in \{c, C_F, C_R, C_M\}, \\ \textit{ambiguous (numerically +)}, & \theta \in \{\beta, p\}. \end{cases}$$

The signs for $\{R, c, C_F, C_R, C_M\}$ are distribution-free (provable via the FOC comparative statics below); the β sign is not *distribution-free—it is positive numerically ($\geq 7/8$ cells) but not signed in general, consistent with Role 1's finding that the carbon direction is governed by the distributional functional $\Phi(F)$ (carbon falls iff $\Phi(F) > 0$).*

Proof sketch. Applying the envelope theorem to Π_{FCL}, $\partial\text{Carbon}_{\text{FCL}}/\partial\theta = c \cdot E[\mathbf{1}_{x+d^* \in (0,K_F^*)} \cdot \partial(x + d^*)/\partial\theta] + c\left(1 - F(K_F^*)\right)\partial K_F^*/\partial\theta$ (since $\partial_{K_F} E[\min(x_+, K_F)] = \Pr(x > K_F) = 1 - F(K_F)$; and similarly for no-FCL). Differentiating the two FOCs gives $\partial K_F^*/\partial R = \bar{q}_F/(f(\beta + p - R)) > 0$, $\partial K_F^*/\partial c < 0$, and $\partial K_M^*/\partial C_M < 0$. Substituting into the difference J collapses the ambiguous terms because the no-FCL problem is independent of R, C_M; the remaining sign follows from the mining substitution term $R \cdot d^*$. The full case-by-case expansion is deferred to the appendix; the numerical companion (`characterization_comparative_statics.csv`) confirms the distribution-free signs in every cell and the β sign in $\geq 7/8$ of the tested distributions. □

Boundary movement

The adoption threshold $\bar{C}_M = (R-c)(\beta+p-c-C_F)/(\beta+p-c)$ satisfies $\partial\bar{C}_M/\partial R > 0$, $\partial\bar{C}_M/\partial c < 0$, $\partial\bar{C}_M/\partial C_F < 0$; the numerical run confirms $\partial\bar{C}_M/\partial R \approx +0.96$, $\partial\bar{C}_M/\partial c \approx -0.97$, $\partial\bar{C}_M/\partial C_F \approx -0.04$.

2 Optimal-contract comparative statics

Fix (K_R^*, K_F^*, K_M^*) and compare the myopic and carbon-optimal dispatch schedules $d_M(x) = \min(\max(K_F^* - x, 0), K_M^*)$ and $d_C(x) = \min(\max(x, 0), K_M^*)$. Define the *wedge* $W = c\,E[\text{fossil}_M - \text{fossil}_C] \geq 0$.

Lemma 1 (Wedge monotonicity, under a regular tail)**.** *For residual-demand laws with a non-increasing shift ratio (regular families; excludes heavy Pareto tails), $\partial W/\partial C_M \le 0$ and $\partial W/\partial C_R \ge 0$.*

Proof. K_M^* is decreasing in C_M (FOC $F(K_F - K_M) = C_M/(R-c)$), which shrinks the support on which $d_M \neq d_C$. The regular-tail hypothesis enters here: with a non-increasing shift ratio the wedge integrand is single-peaked in x, so shrinking the mismatch support strictly lowers W; without it (heavy Pareto tail) mass can migrate into the tail of the mismatch region faster than the support shrinks, and the sign can flip. Conversely, higher C_R lowers K_R^*, shifts residual demand rightward, and expands the region on which the myopic rule fails to substitute fossil. □

The numerical run in `contract_wedge_comparative_statics.csv` confirms $\partial W/\partial C_M \le 0$ in 7/8 cells; the lone exception is the heavy-tailed Pareto case, which is exactly why the Lemma carries the regular-tail hypothesis (without it the inequality can fail).

3 Value of robustness

Let $\mathcal{F}(r) = \{\text{Beta}(a_0 + s, b_0 - s) : |s| \le r,\ b_0 - s > 0\}$ be a one-parameter ambiguity set indexed by the *shift bound* r along an MLR path (not a Wasserstein-1 ball: W_1 of the shifted law from F_0 is not $|s|$, and a genuine W_1 set is nonparametric—we make no W_1 claim). The sets nest, $\mathcal{F}(r) \subset \mathcal{F}(r')$ for $r < r'$. Define the robust joint problem $\max_K \min_{F \in \mathcal{F}(r)} \Pi(K; F)$.

Theorem 1 (Price of robustness is monotone)**.** *On the Beta family, the price of robustness $\rho(r) := \Pi(K^*(F_0); F_0) - \max_K \min_{F \in \mathcal{F}(r)} \Pi(K; F)$ is nondecreasing in r.*

Proof. Monotonicity of ρ follows from set inclusion: for $r < r'$, $\mathcal{F}(r) \subset \mathcal{F}(r')$, so the inner minimum is weakly smaller and the robust value weakly worse, hence $\rho(r) \le \rho(r')$. □

(Numerical observation.) Along the MLR path the carbon gap $\Delta_c(r) := \text{Carbon}_{\text{robust}} - \text{Carbon}_{\text{nominal}}$ keeps a constant sign across the scanned $[0, 1.2]$ grid (`robustness_gap_vs_radius.csv`); this is a numerical finding, not proved (at $r = 0$ robust = nominal so $\Delta_c(0) = 0$, and there is no sign to "inherit").

4 Fragility map

Proposition 2 (Adoption boundary: a kink, not a jump)**.** *As $C_M \uparrow \bar{C}_M$, the K_M FOC $F(K_F^* - K_M^*) = C_M/(R-c)$ forces $F(K_F^* - K_M^*) \uparrow \varphi^\circ = F(K_F^\circ)$ while $K_F^* \to K_F^\circ$, so $K_M^* \to 0$* continuously *($K_M^*(\bar{C}_M^-) = 0$). Carbon, profit, and all capacities are therefore* continuous *through $\bar{C}_M$; the adoption boundary is a* kink *(a discontinuity in $\partial K_M^*/\partial C_M$, which drops to 0 for $C_M > \bar{C}_M$), not a jump. There is no discontinuous carbon change at the threshold.*

The numerical witness table `fragility_adoption_boundary.csv` shows K_M^* sliding smoothly from 0.16 to 0 across $\bar{C}_M \pm 0.3$ — a continuous decline over a 0.6-wide window, confirming the kink (no jump).

Proposition 3 (Endpoint shift-ratio boundary)**.** *With the shift ratio $g(x) = F_{K_R}(x)/f(x)$, the capacity channel $K_R^* > K_R^\circ$ is driven by the endpoint inequality $g(K_F^* - K_M^*) \ge g(K_F^*)$ (lower dispatch endpoint carries the larger shift ratio), which holds automatically wherever g is non-increasing (regular families). On the Beta$(a, 5-a)$ family it holds for all $a > a^*$ and fails only where g turns increasing; empirically $a^* \approx 0.5$.*

The two witnesses on either side of a^* are recorded in `fragility_endpoint_boundary.csv`.

Summary of verified claims

- Proposition **??**: 5/7 primitives have a distribution-free sign (all except β and p; β is positive numerically but not distribution-free, p is small-FCL sensitive).
- Lemma **??**: confirmed in 7/8 (C_M) and 8/8 (C_R) distributions (the C_M exception is the heavy-tail Pareto case the regular-tail hypothesis excludes).
- Theorem **??**: price of robustness *monotone* in r (proved by set inclusion); separately, the LR-path carbon-gap sign invariance holds numerically (a numerical observation, not a theorem), with price of robustness ≤ 0.125 in the tested grid.
- Propositions **??** and **??**: explicit witnesses supplied.